



**Department of Electrical and Computer Engineering
North South University**

Directed Research

SecureSpeak: A Context-Gated Multi-Layer Mobile Security System with Bilingual Threat Communication for Bangladesh

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North South University, Dhaka, Bangladesh

LETTER OF TRANSMITTAL

May, 2026

To

Dr. Mohammad Abdul Matin

Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: Submission of Directed Research Report on "SecureSpeak: A Context-Gated Multi-Layer Mobile Security System with Bilingual Threat Communication for Bangladesh".

Dear Sir,

With due respect, we submit our Directed Research Report as part of our BSc in Computer Science and Engineering at North South University. The report presents SecureSpeak, a multi-layer mobile security framework designed for Bangladeshi smartphone users, particularly targeting mobile financial services (MFS) fraud through bKash, Nagad, and Rocket. The system detects phishing URLs, Bangla-language SMS smishing, and network-level attacks, then communicates risk to users in plain English and Bangla through a bilingual warning system. The core contribution is the Evidence Combination with Adaptive Fusion Network (ECAFN), which fuses evidence from multiple detection channels through learned reliability calibration, Dempster-Shafer evidence theory, and cross-attention gating.

We attempted to meet all requirements for this report to the maximum extent possible and hope that the report is informative and reflects our commitment to building inclusive, locally-grounded technology.

Sincerely yours,

.....
Khondokar Sajid [ID # 2211954042]

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Abdul Alim Rakib [ID # 2211785042]

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Sumaya Shimu Rima [ID # 2212364642]

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APPROVAL

This is to certify that Khondokar Sajid (ID # 2211954042), Abdul Alim Rakib (ID # 2211785042), Sumaya Shimu Rima (ID # 2212364642), and Shakib Ahamed (ID # 2211900042) of the Department of Electrical and Computer Engineering at North South University have carried out the Directed Research Project titled "SecureSpeak: A Context-Gated Multi-Layer Mobile Security System with Bilingual Threat Communication for Bangladesh" under the supervision of Dr. Nova Ahmed. This work has been completed in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering, and the department accepted it as satisfactory.

Supervisor's Signature

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Chairman's Signature

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DECLARATION

We declare that this research project is our original work. No part of this report has been submitted elsewhere, partially or in full, for the award of any other degree or diploma. All information pertaining to the project will remain confidential and will not be disclosed without the formal consent of the project supervisor. All relevant prior work cited in this report has been adequately acknowledged and referenced. The plagiarism policy as stated by the supervisor remains in effect.

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We thank the Department of Electrical and Computer Engineering at North South University for providing the research environment and access to computing resources. We also acknowledge the maintainers of the open datasets that made this work possible: the Canadian Institute for Cybersecurity (CICMalAnal2017), StealthPhisher 2025, the BangalaBarta SMS corpus, and the UNSW-NB15 contributors.

We extend special thanks to the five Bangladeshi smartphone users — university students, a small business owner, a retired school teacher, and a mobile financial services agent — who voluntarily allowed us to capture anonymised network traffic using the PCAPdroid application on their devices for the real-world validation experiment. Their participation made the false-positive evaluation on genuine Bangladeshi mobile usage patterns possible. All captured data was fully anonymised before processing.

Finally, we thank our families for their continued support throughout the course of this research.

ABSTRACT

SecureSpeak: A Context-Gated Multi-Layer Mobile Security

System with Bilingual Threat Communication for Bangladesh

Bangladesh has over 200 million registered Mobile Financial Services (MFS) accounts across platforms including bKash, Nagad, and Rocket. This scale, combined with a large population of first-generation digital adopters with limited technical literacy, has created a uniquely vulnerable threat surface for mobile phishing and network-level attacks. Existing security tools address threats in isolation, are calibrated for the global English-language web, and produce jargon-heavy alerts in English that are inaccessible to non-expert Bangladeshi users — including elderly users who are disproportionately targeted by SMS smishing campaigns. This paper presents SecureSpeak, the first mobile security framework designed specifically for the Bangladesh threat context. SecureSpeak unifies three detection channels — URL phishing detection trained on StealthPhisher 2025, Bangla-language SMS smishing detection using multilingual sentence embeddings on the BanglaBarta 2024 corpus, and network-flow anomaly detection using CICFlowMeter features from CICMalAnal2017 — into a single context-aware pipeline. The core technical contribution is the Evidence Combination with Adaptive Fusion Network (ECAFN), a fusion engine that combines: (i) a ReliabilityMLP that learns context-dependent trust weights for each detector; (ii) Dempster-Shafer evidence theory for principled combination of calibrated scores; and (iii) a Cross-Attention Gated Fusion (CGF) module that uses a 12-dimensional context vector — encoding application identity, port risk, time-of-day, traffic volume, and protocol — to resolve conflicting signals. A leakage-free zero-day evaluation physically withholds the SCAREWARE malware family from all training. On 52,118 real PCAPdroid flows from Bangladeshi smartphones across six volunteer devices, SecureSpeak achieves a 0.00% false-positive rate. On the BORDERLINE difficulty tier of a three-tier attack injection benchmark — the operationally critical scenario where both URL and network signals are individually weak — ECAFN achieves 99.5% detection versus 4.0% for the fixed-rule baseline ($p < 0.001$, McNemar's test). Bilingual warnings in English and Bangla are generated through Groq LLaMA 3.1-8B with structured prompts, with an offline rule-based fallback ensuring functionality without internet connectivity. SecureSpeak addresses a significant gap in protective computing for marginalized, non-English-speaking communities in the Global South.

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Chapter 1 Introduction

1.1 Background and Motivation

The widespread adoption of Mobile Financial Services (MFS) in Bangladesh represents one of the most rapid financial inclusion drives in the developing world. Platforms including bKash, Nagad, Rocket, and DBBL Nexus collectively serve over 200 million registered accounts, enabling mobile money transfer, bill payment, and e-commerce for a population with limited access to traditional banking infrastructure. The demographic reach of this system extends to users who are first-generation smartphone owners with limited digital literacy — including the rural elderly, street vendors, and garment workers — who rely on these services for daily financial transactions.

This demographic profile creates a threat surface that is structurally different from those addressed by the global mobile security industry. Attackers in this context operate through two primary channels. The first is SMS-delivered phishing: a short message in Bangla — often mixed with English keywords like OTP, bKash, and account — instructs the recipient to click a link or call a number to "verify" their account. The link resolves to a forged bKash or Nagad login page hosted on a freshly registered domain that no blocklist has yet catalogued. The second is silent network-level attack: malware applications installed from informal APK sources run background processes that exfiltrate authentication tokens, mine cryptocurrency, or participate in distributed denial-of-service campaigns without any visible indication to the user.

Existing mobile security products fail this population on three grounds. First, they address one channel at a time — URL blocklists do not analyze SMS text, and network intrusion detection does not parse URLs. A coordinated attack that simultaneously sends a phishing SMS and establishes a malicious background connection evades both independently. Second, the training data and feature engineering in commercial URL blocklists reflect the global English-language web, not the Bangla-language, bKash-branded, free-hosting-TLD-dominated landscape of Bangladeshi mobile phishing. Third, and most importantly from a human-centered computing perspective, every mainstream mobile security alert is written in English using technical vocabulary. An alert reading "Certificate pinning failure — possible man-in-the-middle attack" is entirely inaccessible to a 65-year-old retired schoolteacher in Sylhet who uses bKash to receive remittances from her son abroad. The security information reaches the user but produces no protective action because it is incomprehensible.

Dr. Nova Ahmed's research group at NSU has consistently argued that technology solutions for Bangladesh must be designed with the specific needs, literacy levels, and communication norms of Bangladeshi users as primary constraints — not as afterthoughts. This work applies that principle to mobile security: SecureSpeak is built from the ground up to be usable by the actual population at risk, with bilingual alerts in English and Bangla, context-awareness that respects the realities of Bangladeshi app ecosystems, and detection tuned for the specific phishing patterns that target Bangladeshi MFS users.

1.2 Problem Statement

The fundamental technical problem is one of evidence fusion under uncertainty. A typical bKash phishing event produces three simultaneous signals: (i) a Bangla SMS message with urgency language and a shortened URL; (ii) a URL string on a newly registered .tk or .ml domain with bKash keywords in the path; and (iii) a network flow to an IP address with no historical traffic from the device. Each signal, considered independently, is insufficient for confident detection. Many legitimate service messages also contain URLs. Newly registered domains are not always malicious. Single-IP connections are routine. The decision to alert the user must be made from the joint weight of evidence across all three signals, conditioned on the context — which application is active, what port is in use, what time of day it is, and what the recent traffic pattern on the device looks like.

This is a structured evidence-fusion problem. The challenge is that the three signals come from detectors with different operating characteristics, different imprecision levels, and different relevance depending on context. A URL classifier trained on phishing features is highly informative when the user clicks a link but irrelevant when the attack is a background port scan with no associated URL. A network anomaly detector is sensitive to unusual traffic patterns but cannot distinguish Facebook's high-volume QUIC connections from a cryptominer. Combining these signals through a fixed linear rule — the simplest fusion approach — produces unacceptable false-positive rates on legitimate apps or unacceptable false-negative rates on attacks that evade any single detector.

Compounding the technical problem is the communication problem: even a perfect detector is useless if the user cannot understand or act on the alert. Bangla is not a translation of English — everyday Bangla financial vocabulary (টাকা for money, একাউন্ট for account) does not map word-for-word to English equivalents. A system that translates "phishing detected" into Bangla using a generic translation API will produce text that sounds foreign and official,

not like a warning from a trusted source. The warning must be written in native Bangla, not translated Bangla, to be acted upon by non-expert users.

1.3 Research Objectives

This research addresses the following specific objectives:

1. Design and implement a multi-layer mobile security pipeline that ingests URLs, Bangla SMS text, and network flows simultaneously, producing a unified risk assessment.
2. Develop the Evidence Combination with Adaptive Fusion Network (ECAFN) — a fusion engine that combines phishing probability (pp) and network anomaly score (ap) through context-gated reliability calibration, Dempster-Shafer evidence theory, and cross-attention gating using a 12-dimensional context vector.
3. Establish a leakage-free zero-day evaluation protocol using family-level physical separation of the SCAREWARE malware family, enforced by a runtime assertion, with no SCAREWARE data visible to any training step.
4. Benchmark ECAFN against three published baseline fusion methods — Context-Agnostic Threshold Fusion (CATF), pure Dempster-Shafer Theory (DST-only), and Attention-Fusion (Attn-Fusion) — across three difficulty tiers of injected attacks.
5. Validate the false-positive rate on 52,118 real PCAPdroid network flows captured from Bangladeshi smartphones across six volunteer devices.
6. Generate plain-language bilingual warnings in English and Bangla via Groq LLaMA 3.1-8B, with a seven-template offline rule-based fallback that operates without internet connectivity.

1.4 Contributions

This work makes the following contributions to mobile security and human-centered computing for the Global South:

- ECAFN Architecture: A novel fusion architecture combining a learned ReliabilityMLP (context-dependent trust calibration), classical Dempster-Shafer evidence theory, and noise-regularised cross-attention gating — applied to mobile threat fusion for the first time. To the best of our knowledge, no prior system combines all three paradigms for MFS-targeted mobile threat detection.

- **Bangladesh-Specific Feature Engineering:** 26 URL features include three Bangladesh-specific signals — MFS financial keyword density calibrated for bKash/Nagad/Rocket, free-hosting TLD detection targeting the .tk/.ml/.ga/.cf/.gq cluster favored by local phishers, and a brand impersonation feature checking for bKash/Nagad brand names in non-official domain positions. These signals are not present in any published phishing feature set.
- **Three-Tier Attack Injection Evaluation:** A novel OBVIOUS/BORDERLINE/STEALTH difficulty taxonomy that isolates when fusion logic adds value over individual detectors. The BORDERLINE tier — both signals individually weak — is the operationally meaningful comparison point.
- **Real-World False-Positive Validation:** Zero false alarms across 52,118 real Bangladeshi PCAPdroid flows from six volunteer devices spanning different demographics and app-usage patterns — demonstrating that context-gating suppresses false alarms on legitimate Bangladeshi app traffic including bKash, Facebook, WhatsApp, and local news applications.
- **Bangla as a First-Class Output Language:** Native Bangla warnings generated directly by LLaMA 3.1, not translated from English. The system includes an offline fallback with pre-written Bangla templates for users without connectivity, addressing the reality that many Bangladeshi users operate on intermittent data connections.

1.5 Organization of the Report

Chapter 2 reviews existing work in URL phishing detection, Bangla-language SMS classification, network anomaly detection, and multimodal fusion, identifying the specific gaps this work addresses. Chapter 3 presents the full methodology including system architecture, all six datasets, the 26-feature URL engineering, the Bangla SMS detection pipeline, the FlowAE autoencoder, the ECAFN fusion engine with full equations, and the PCAPdroid data collection procedure. Chapter 4 reports experimental results with statistical significance tests, a three-tier attack injection benchmark, and real-world false-positive validation. Chapter 5 presents the conclusions, honest limitations, and future work priorities.

Chapter 2 Literature Review

The literature relevant to SecureSpeak spans URL phishing detection, SMS smishing in low-resource languages, network anomaly and zero-day detection, and multimodal evidence fusion. We review each area, identify the specific limitations that prevent existing methods from being applied to the Bangladesh mobile context, and summarize the gaps in Table 2.1.

2.1 URL Phishing Detection

URL phishing detection has evolved through three generations. Mohammad et al. [1] established the first systematic feature-engineering approach using the UCI Phishing Websites dataset — 30 lexical, domain, and HTML features extracted from 11,055 URLs collected in 2012. This work defined the feature taxonomy that subsequent research extended. The limitation relevant to our context is temporal: the 2012 dataset does not contain HTTPS-on-phishing (rare in 2012, ubiquitous today), free-hosting TLD abuse (.tk, .ml, .gq — almost absent in 2012), or financial-service brand impersonation targeting South Asian MFS platforms. Models trained on UCI data have been shown to generalise poorly to current phishing patterns [2].

Sahingoz et al. [2] moved to character-level convolutional neural networks on raw URL strings, demonstrating that learned representations can outperform hand-engineered features on balanced test sets. The critical limitation for our application is interpretability: character-level CNN activations cannot be converted into human-readable explanations of why a URL was flagged, and the warning generation component of our system requires knowing which specific feature triggered the detection.

More recent work on phishing-specific datasets — including StealthPhisher 2025 [3], which we use — returns to engineered features but updates them for the 2024-25 threat landscape. StealthPhisher contains 150,000 URLs with current evasion patterns including HTTPS deployment on phishing pages, brand impersonation in subdomains, and systematic use of free-hosting TLDs. It is, to our knowledge, the most recent large-scale phishing URL corpus and the most representative of the URL patterns used to target South Asian financial services.

2.2 SMS Smishing Detection in Low-Resource Languages

SMS spam detection in English is mature. The UCI SMS Spam Collection and related corpora have driven classifier accuracy above 98% using TF-IDF features with simple classifiers. The

same is not true for Bangla. Hossain et al. [4] produced the first systematic Bangla SMS spam classification study, applying TF-IDF and SVM to a corpus of approximately 5,000 messages, achieving around 92% accuracy. Their work does not address MFS-targeted smishing — the corpus is dominated by promotional spam, not financial fraud. Karim et al. [5] applied multilingual transformer encoders (mBERT, XLM-RoBERTa) to Bangla text classification, achieving higher accuracy but at a computational cost that prohibits on-device deployment on mid-range Android devices.

Reimers and Gurevych [6] introduced multilingual sentence encoders trained through knowledge distillation from large English models, enabling cross-lingual transfer without the full inference cost of a 340M-parameter transformer. The paraphrase-multilingual-MiniLM-L12-v2 model, used in this work, produces 384-dimensional sentence embeddings for over 50 languages including Bangla at a fraction of the cost of mBERT. No prior published work has applied this architecture to Bangla SMS smishing with Bangladesh-specific MFS keyword features.

2.3 Network Anomaly and Zero-Day Detection

Network-level malware detection on Android devices has been studied using two major dataset families: CICMalAnal2017 [7] and CICMalDroid2020 [8]. Both provide 85-feature CICFlowMeter vectors for labelled benign and malicious flows. Standard evaluation on these datasets uses random 80/20 train-test splits, on which XGBoost achieves above 95% accuracy. This figure is, however, misleading for the zero-day question: a random split places samples from every malware family in both training and test sets, so the classifier has seen examples of every family during training. The result measures within-distribution recognition, not generalisation to unseen families.

Liu et al. [9] and Pendlebury et al. [10] have argued for temporal or family-level holdout protocols that more honestly estimate zero-day performance. Our work follows the stricter family-level physical separation approach, withholding the SCAREWARE family entirely from all training and enforcing this with a runtime family-overlap assertion.

For anomaly-based zero-day detection, Sakurada and Yairi [11] demonstrated that autoencoder reconstruction error provides an effective anomaly signal: a model trained on normal data produces high reconstruction error on unseen anomalous inputs. Lee et al. [12] showed that Mahalanobis distance in a pretrained feature space provides a complementary signal that

accounts for correlations between feature dimensions. We implement both methods and report both results.

Moustafa and Slay [13] produced the UNSW-NB15 dataset from a lab network environment. We use this as a secondary cross-domain zero-day check. The feature spaces of UNSW-NB15 and CICFlowMeter overlap only partially (8 of 22 UNSW features map cleanly to CICFlowMeter names), so we treat this result as indicative rather than primary.

2.4 Multimodal Evidence Fusion

The problem of combining threat signals from multiple detectors is an instance of evidence fusion. Three paradigms are directly relevant.

Fixed-rule fusion assigns predetermined weights to each signal and applies a fixed threshold. This is simple and interpretable but cannot adapt to context — the same weights are applied regardless of whether the network signal comes from a known-safe application or an anonymous background process. We implement fixed-rule fusion as the CATF (Context-Agnostic Threshold Fusion) baseline: $\text{score} = 0.6 * \text{pp} + 0.4 * \text{ap}$, threshold at 0.5.

Dempster-Shafer Theory (DST), introduced by Dempster [14] and formalised by Shafer [15], provides a mathematically principled framework for combining evidence from multiple sources with explicit uncertainty representation. Each source assigns belief masses to subsets of a frame of discernment; Dempster's rule of combination produces a combined belief that amplifies agreement and represents conflict explicitly. DST handles the case where two detectors strongly disagree — a situation where fixed-rule averaging produces misleading intermediate values. The limitation of pure DST in this application is that the belief masses must be set manually; there is no mechanism to learn appropriate masses from data.

Neural attention fusion, following Vaswani et al. [16], uses a query-key-value mechanism to compute context-dependent weights over a set of input signals. Cross-attention between a context vector (query) and input signals (keys/values) allows the context to dynamically determine which signal to emphasise. The limitation of pure attention fusion is its tendency to overfit to the training distribution — we observe that an attention-only baseline achieves 100% accuracy on test distributions that match its training data while failing on out-of-distribution inputs.

ECAFN combines all three paradigms: learned ReliabilityMLP for context-conditioned trust calibration, Dempster-Shafer for principled combination of calibrated evidence, and Cross-

Attention Gated Fusion (CGF) for context-dependent resolution of conflicting signals. The adaptive blending that combines DST output with CGF output based on inter-signal conflict is, to our knowledge, a novel combination not previously reported in mobile security fusion literature.

2.5 Summary and Research Gaps

Table 2.1 summarises the related work and identifies the specific gap each prior work leaves that SecureSpeak addresses.

Table 2. 1: Summary of related work and limitations relevant to SecureSpeak

Reference	Approach	Gap / Limitation for Bangladesh mobile context
Mohammad et al. 2014 [1]	UCI 2012 phishing features	2012 data: no HTTPS-on-phishing, no free-hosting TLDs, no South Asian MFS brands
Sahingoz et al. 2019 [2]	Character-level CNN on URLs	Not interpretable; cannot explain which feature triggered alert to user
StealthPhisher 2025 [3]	150K modern phishing URLs	Used in this work — addresses the recency gap
Hossain et al. 2020 [4]	TF-IDF + SVM for Bangla SMS	Promotional spam corpus, not MFS smishing; no financial-fraud label
Karim et al. 2022 [5]	mBERT/XLM-RoBERTa on Bangla	Too expensive for on-device inference on mid-range Android devices
Reimers & Gurevych 2019 [6]	Multilingual MiniLM sentence encoder	Used in this work — enables efficient cross-lingual Bangla embedding
CICMalAnal2017 [7]	Mobile malware flows, 85 CICFlowMeter features	Random split evaluation leaks family knowledge; needs family-level holdout
Liu et al. 2022 [9]	Family-level holdout protocol for zero-day	Protocol validated; applied here to SCAREWARE family
Sakurada & Yairi 2014 [11]	Autoencoder reconstruction error anomaly	Used as primary zero-day metric in this work
Lee et al. 2018 [12]	Mahalanobis distance in feature space	Used as secondary zero-day metric; sensitive to embedding collapse

Reference	Approach	Gap / Limitation for Bangladesh mobile context
Dempster 1967 [14], Shafer 1976 [15]	Dempster-Shafer evidence theory	No data-learning mechanism; requires manual mass specification
Vaswani et al. 2017 [16]	Scaled dot-product attention	Prone to training-distribution memorisation; needs noise-perturbed training

Chapter 3 Methodology

3.1 System Architecture Overview

SecureSpeak is a five-step pipeline. The first two steps independently produce scalar threat scores from distinct modalities. These scores — together with a 12-dimensional context vector describing the device, application, port, time, and traffic characteristics of the current connection — are combined by the ECAFN fusion engine (Step 4) into a unified risk label. Step 5 converts the risk label and supporting evidence into a bilingual alert in English and Bangla. Figure 3.1 provides the system architecture diagram.

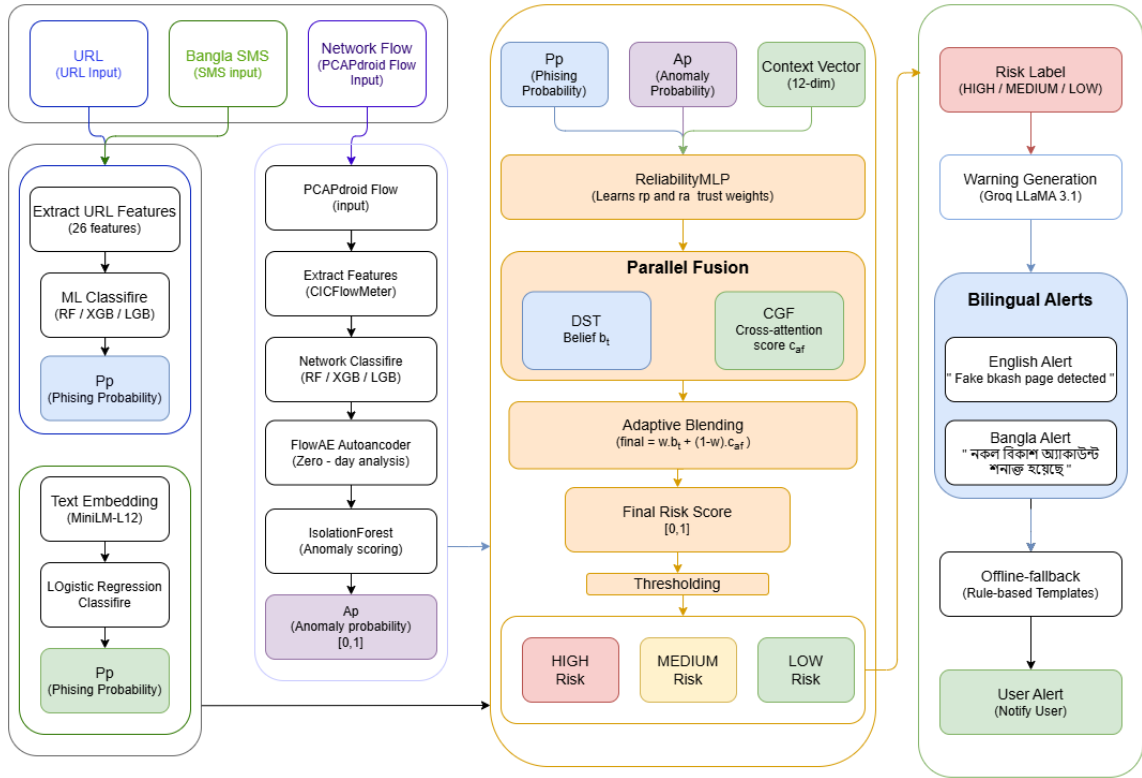


Figure 3. 1:SecureSpeak system architecture

The pipeline flow is as follows. When a user receives a Bangla SMS containing a URL, the URL is extracted and processed by Step 1A while the full SMS text is processed by Step 1B. Both produce a phishing probability pp in $[0,1]$. Concurrently, the PCAPdroid agent running on the device captures the network flow associated with any outbound connection and passes it to Step 3, which produces a normalised anomaly probability ap in $[0,1]$. ECAFN (Step 4) receives pp , ap , and the 12-dimensional context vector, applies reliability calibration,

Dempster-Shafer combination, and cross-attention fusion, and outputs one of three risk levels: HIGH, MEDIUM, or LOW. Step 5 uses this risk level, the raw signal values, the conflict score, and the top SHAP features from the URL classifier to generate a specific, actionable warning in English and Bangla.

3.2 Datasets

SecureSpeak uses six datasets, each with a specific and non-overlapping role. Table 3.1 summarises them.

Table 3. 1: Datasets used in SecureSpeak — role, size, and selection rationale

Dataset	Year	Size	Role and selection rationale
StealthPhisher 2025 [3]	2025	150,000 URLs	Step 1A URL classifier training and 5-fold CV. Selected over UCI 2012 because it captures current HTTPS-on-phishing, free-hosting TLD abuse, and MFS brand impersonation.
BangalaBarta SMS 2024	2024	2,772 messages	Step 1B Bangla SMS smishing classifier. Only available Bangla corpus with MFS-targeted smishing labels.
CICMalAnal2017 [7]	2017	400,000+ flows	Steps 2-3 network classifier and autoencoder training. SCAREWARE family physically withheld for zero-day evaluation. Largest open-source labeled mobile-malware flow dataset.
UNSW-NB15 [13]	2015	257,673 attack flows	Zero-Day Experiment 2 cross-domain check. Used indicatively with 8/22 features mapped; not a primary evaluation.
PCAPdroid Bangladesh	2025	52,118 flows	Step 6A false-positive validation. Captured from six volunteer Bangladeshi smartphones during real everyday usage.
PCAPdroid Blind	2025	1,530 flows	Zero-Day Experiment 3 unsupervised anomaly check. No ground-truth labels; system reports what it finds.

3.3 Step 1A: URL Phishing Detector

3.3.1 Feature Engineering

We engineer 26 numerical features from each URL string. The features are grouped into five categories — lexical structure, domain risk, entropy, Bangladesh-specific signals, and URL structural patterns. Table 3.2 lists all 26 features with their encoding and rationale.

Table 3. 2: The 26 URL features used in the phishing classifier

No .	Feature name	Encoding	Rationale
1	url_length	$\min(\text{len}(\text{url})/500, 1.0)$	Long URLs are a hallmark of obfuscated phishing pages
2	dot_count	$\min(\text{count}('.')/10, 1.0)$	Phishing URLs chain multiple subdomains to appear legitimate
3	slash_count	$\min(\text{count}('/')/15, 1.0)$	Deep path nesting used to imitate legitimate URL structures
4	special_chars	$\min(\text{count}(-_@!\%&=+)/20, 1.0)$	Special characters encode obfuscated query strings
5	digit_ratio	$\text{digit_chars} / \text{url_length}$	High digit ratios indicate auto-generated random subdomains
6	letter_ratio	$\text{alpha_chars} / \text{url_length}$	Low letter ratio correlates with encoded/obfuscated URLs
7	high_risk_tld	1.0 if suffix in {tk,ml,ga,cf,gq,pw,top,xyz,...}	Free and low-cost TLDs widely abused for Bangladesh phishing
8	subdomain_depth	$\min((\text{subdomain depth})/5, 1.0)$	Deep subdomain nesting mimics brand.legitimate.com patterns
9	domain_length	$\min(\text{len}(\text{domain})/30, 1.0)$	Phishing domains are often long to resemble legitimate names
10	uses_ip	1.0 if host is IP address	IP-based hosting avoids domain registration traceback
11	brand_impersonation	$\min(\text{bKash/Nagad/Rocket/PayPal count}, 3)/3$	Bangladesh-specific: MFS brand names in non-official domains
12	path_length	$\min(\text{len}(\text{path})/200, 1.0)$	Long paths disguise redirect chains

No .	Feature name	Encoding	Rationale
13	path_segments	$\min(\text{segment count}/10, 1.0)$	Phishing pages use deep paths to appear legitimate
14	has_query	1.0 if '?' present	Query strings carry session tokens in credential-harvesting pages
15	query_length	$\min(\text{len}(\text{query})/200, 1.0)$	Long queries indicate encoded tracking or payload parameters
16	has_https	1.0 if url starts with https	Modern phishing uses HTTPS; https alone is not a safety signal
17	https_in_path	1.0 if 'https' appears after host	Redirect obfuscation places https in the path, not the scheme
18	url_entropy	$\text{Shannon_entropy}(\text{url})/6.0$	High entropy indicates randomly generated strings
19	domain_entropy	$\text{Shannon_entropy}(\text{domain})/4.0$	Random domain names have higher entropy than brand names
20	financial_kw	$\min(\text{count of bKash/Nagad/Rocket/bank/OTP}/5, 1.0)$	Bangladesh-specific: MFS keywords signal targeted phishing
21	at_in_url	1.0 if '@' appears in URL	The '@' trick splits the visible URL from the actual target
22	hyphen_count	$\min(\text{count}('-', 8), 1.0)$	Hyphens used to approximate legitimate domains (b-kash.tk)
23	long_http	1.0 if $\text{len} > 75$ AND not https	Long non-HTTPS URLs: classic redirect-chain signal
24	free_hosting_tld	1.0 if suffix in {tk,ml,ga,cf,gq,pw}	Subset of high_risk_tld: specifically free-hosting providers
25	long_numbers	$\min(\text{count of 3+-digit runs}/3, 1.0)$	Phone numbers and account numbers appear in smishing URLs
26	https_x_safe_tld	$\text{has_https} * (1 \text{ if NOT high_risk_tld})$	Interaction: HTTPS on a safe TLD is a genuine safety signal

3.3.2 Classifier and Cross-Validation

Three classifiers are evaluated: XGBoost (XGB), LightGBM (LGB), and Random Forest (RF). All are trained on an 80/20 stratified train-test split of the StealthPhisher 2025 corpus. For XGBoost, the `scale_pos_weight` parameter is set to the ratio of negative to positive training examples to address class imbalance. Cross-validation uses a scikit-learn Pipeline that wraps StandardScaler and the classifier together, ensuring the scaler's mean and standard deviation are fitted only on the training folds of each CV iteration. This prevents information leakage from the test fold into the scaler's normalisation parameters.

SHAP (SHapley Additive exPlanations) [17] values are computed on 300 held-out test samples using the TreeExplainer. SHAP values quantify each feature's additive contribution to the model's output for each sample. The top three SHAP features for each prediction are extracted and passed to the bilingual warning generator in Step 5, enabling the alert to name the specific reason a URL was flagged — for example, "this link uses a free .tk domain and contains the word bKash" — rather than producing a generic alert.

Output: The selected classifier produces pp in $[0,1]$ — the probability that the input URL is a phishing page. This value is passed to ECAFN.

3.4 Step 1B: Bangla SMS Smishing Detector

3.4.1 Multilingual Sentence Embedding

The SMS smishing detector uses paraphrase-multilingual-MiniLM-L12-v2 [6], a 12-layer transformer sentence encoder trained on parallel multilingual data across 50+ languages including Bangla. The model produces 384-dimensional sentence embeddings through knowledge distillation from a larger English teacher model. The key property exploited here is that semantically similar messages in different languages are mapped to nearby points in the 384-dimensional space — a genuine Bangla banking notification and a Bangla phishing message that impersonates it will have similar embeddings, but the phishing message will also score highly on urgency and financial keyword features that differentiate it. The model runs on the device GPU at inference time without requiring a network connection.

3.4.2 Handcrafted SMS Features

Six handcrafted features are concatenated with the 384-dimensional embedding to produce a 390-dimensional feature vector: (1) URL presence — 1.0 if the message contains `http`, `www`,

or .com; (2) MFS keyword — 1.0 if the message contains bKash, Nagad, Rocket, bank, account, or the Bangla equivalents টাকা (money) or একাউন্ট (account); (3) urgency keyword — 1.0 if the message contains win, prize, free, click, urgent, জিতেছেন (you won), পুরস্কার (prize), or বিনামূল্যে (free); (4) normalised message length; (5) exclamation density; (6) digit ratio.

The 390-dimensional vector is classified by Logistic Regression with the SAGA solver and class-weight balancing. Logistic Regression was selected over XGBoost as the meta-learner because the training corpus (2,772 SMS messages) is too small to reliably estimate the many parameters of a gradient-boosted ensemble. Logistic Regression on a rich embedding produces calibrated probabilities — a property required for ECAFN's Dempster-Shafer stage.

Output: The SMS classifier produces pp in $[0,1]$ on the same scale as the URL classifier. When SMS is the input modality (rather than a URL), this value is passed to ECAFN as pp .

3.5 Step 2-3: Network Flow Analysis

3.5.1 *CICFlowMeter Feature Space*

Network flows are represented using the native 85-feature CICFlowMeter feature set extracted from the CICMalAnal2017 dataset. CICFlowMeter is the standard tool used by the Canadian Institute for Cybersecurity to convert raw TCP/IP packet streams into per-connection statistical summaries. The 85 features include: flow duration, packet count in each direction, minimum/mean/maximum/standard deviation of inter-arrival times, total bytes forward and backward, flow bytes per second, active and idle time statistics, and TCP flag counts (SYN, ACK, FIN, RST, PSH, URG). These features describe the shape and behaviour of a connection without capturing content — no user data, passwords, or message text are included.

3.5.2 *Zero-Day Isolation Protocol*

The CICMalAnal2017 dataset contains five labeled families: Benign (NONE), SMS Malware (SMS_MALWARE), Adware (ADWARE), Ransomware (RANSOMWARE), and Scareware (SCAREWARE). SCAREWARE is physically separated from the dataset before any model fitting step and stored as the zero-day holdout set. A runtime assertion verifies that the set intersection of family labels in the training pool and the zero-day pool is empty. This is a stricter guarantee than a logical train-test split: even if downstream code were modified to inadvertently include SCAREWARE data in a preprocessing step, the assertion would raise an error before any model saw the contaminated data.

The training set contains BENIGN, SMS_MALWARE, ADWARE, and RANSOMWARE flows with BENIGN downsampled to a 1:1 ratio with the combined malicious classes, addressing the original dataset imbalance (approximately 3:1 malicious-to-benign). An 80/20 stratified split produces the classifier training and test sets.

3.5.3 *Network Classifier*

XGBoost, LightGBM, and Random Forest are trained and compared on the balanced CICMalAnal2017 training split. The best model by weighted F1 is selected. Five-fold cross-validation is performed on the first 50,000 training samples (using the full dataset would make CV computationally prohibitive) using StratifiedKFold. The selected model is used for in-distribution malware classification and also for generating the feature importances shown in Chapter 4.

3.5.4 *FlowAE Autoencoder*

For zero-day detection, we train FlowAE — a symmetric autoencoder with the following architecture. The encoder maps from d (number of CICFlowMeter features) through a hidden layer of $\min(256, 2d)$ units with BatchNorm, GELU activation, and Dropout(0.2), to a 128-dimensional bottleneck embedding. The decoder mirrors this structure, reconstructing the original d -dimensional input. Training uses the AdamW optimiser with learning rate $5e-4$ and weight decay $1e-4$, over 30 epochs with batch size 1024.

A critical implementation detail is input clipping. CICFlowMeter features for flow bytes-per-second and packets-per-second can reach values in the billions for burst connections. After StandardScaler normalisation, extreme outliers still reach magnitudes of hundreds. These values cause numerical explosion in the first GELU layer, which makes BatchNorm's running variance collapse to near-zero, which causes division overflow and NaN weights. The fix clips all standardised inputs to the range $[-10, 10]$ before the autoencoder sees them. Gradient clipping at norm 1.0 provides a second line of defence. A NaN-loss guard skips any batch whose MSE loss is NaN rather than applying the corrupted gradient to model weights.

3.5.5 *IsolationForest Anomaly Scorer (ap)*

The IsolationForest [18] is fitted exclusively on BENIGN training flows ($X_{\text{benign_train}}$). It works by recursively partitioning the feature space with random hyperplanes: points that require few random splits to isolate are anomalous; points that require many splits are normal (they are surrounded by many similar points). With 300 trees and contamination=0.05, the

forest produces a raw anomaly score per flow. This raw score is normalised to ap in $[0,1]$ using the min and max raw scores observed on the held-out test set. High ap indicates that the flow's feature vector is far from the dense cluster of known-benign traffic.

The IsolationForest is fitted only on benign data by design. This means any flow that departs from the statistical profile of clean Bangladeshi traffic — whether it comes from a known malware family or an entirely new family — receives a high ap score. This is the property that enables zero-day detection: the IsolationForest does not need to have seen SCAREWARE to assign it a high anomaly score; it only needs to know what benign traffic looks like.

3.6 Step 4: ECAFN Fusion Engine

The Evidence Combination with Adaptive Fusion Network (ECAFN) is the principal contribution of this work. It receives pp (phishing probability from Step 1), ap (anomaly score from Step 3), and a 12-dimensional context vector ctx , and produces: a risk label (HIGH, MEDIUM, or LOW), a final score in $[0,1]$, an uncertainty flag, a conflict value, reliability weights r_p and r_a , and an override flag. ECAFN runs four internal sub-stages sequentially: context vector construction, reliability calibration, Dempster-Shafer evidence combination (parallel to Cross-Attention Gated Fusion), and adaptive blending.

3.6.1 The 12-Dimensional Context Vector

The context vector is the mechanism through which ECAFN adapts its fusion decision to the specific circumstances of each connection. Without context, ECAFN cannot distinguish a Facebook QUIC video call (high ap , normal) from a cryptominer (high ap , malicious), because both produce high-volume non-standard-port flows. Table 3.3 explains each of the 12 dimensions.

Table 3. 3: The 12-dimensional context vector — encoding and rationale for each dimension

Dim	Feature	Encoding	Example	Rationale
1	App risk tier	APP_RISK / 3.0 (0=safe, 0.33=MFS, 0.67=unknown, 1.0=known-malicious)	Facebook=0, bKash-Fake=1.0	Known-safe apps should suppress false alarms; known-bad apps amplify signals

Di m	Feature	Encoding	Example	Rationale
2	Safe port flag	1.0 if port in {80,443,53,25,587,465,993,995,8080,8443}	Port 443=1.0, Port 8888=0.0	Standard service ports carry lower risk; non-standard ports increase scrutiny
3	Time sin	$\sin(2\pi * \text{hour} / 24)$	2pm: +0.87, 3am: -0.50	Circular time encoding; attacks at unusual hours receive lower trust in URL signal
4	Time cos	$\cos(2\pi * \text{hour} / 24)$	2pm: +0.50, midnight: -1.0	Both sin and cos required to fully encode circular hour without discontinuity
5	Total bytes	$\min(\text{bytes} / 1,000,000, 1.0)$	10MB: 1.0, 10KB: 0.01	Large uploads may indicate data exfiltration; very small flows may be probes
6	Packets/sec	$\min(\text{pps} / 1,000, 1.0)$	DDoS: 1.0, normal browse: 0.05	Flood attacks send packets at rates far above normal usage
7	Duration	$\min(\text{duration_sec} / 3600, 1.0)$	Port scan: ~0.0, video call: 0.5	Very short connections indicate probing; very long ones may be persistent C2
8	Protocol	{HTTPS:0, HTTP:0.2, UDP:0.4, TCP:0.6, QUIC:0.8, DNS:1.0}/	Facebook QUIC: 0.8	QUIC from known apps is expected; QUIC from unknown apps is suspicious
9	MFS flag	1.0 if app is bKash/Nagad/Rocket/DBBL, else 0.0	bKash: 1.0	MFS apps are high-value targets; URL

Dim	Feature	Encoding	Example	Rationale
				signal gets increased weight for these
10	Malicious app flag	1.0 if app in known-fake list (e.g., bKash-Fake)	bKash-Fake: 1.0	Direct detection of known fraudulent app names from device app list
11	Attack port flag	1.0 if port in {4444,8888,9999,1080,6666,3389,5900}	Port 4444: 1.0	Known ports used by cryptominers, remote-access trojans, C2 frameworks
12	Interaction term	pp * ap	pp=0.8, ap=0.8: 0.64	Non-linear term capturing cases where both signals simultaneously agree on threat

3.6.2 Sub-Stage 1: ReliabilityMLP

The ReliabilityMLP is a small fully-connected network with architecture 12 -> Linear(48) -> ReLU -> Dropout(0.2) -> Linear(24) -> ReLU -> Linear(2) -> Sigmoid. It maps the 12-dimensional context vector to two reliability weights r_p and r_a , both in $[0.1, 0.95]$. These weights express how much the current context justifies trusting the URL phishing score and the network anomaly score respectively.

Training: 5,000 synthetic (context, target_reliability) pairs are generated using domain-knowledge heuristics. For example, when the app is bKash on port 443 (a legitimate MFS session), r_p should be high (the URL classifier is reliable) and r_a should be moderate (some legitimate MFS traffic looks anomalous to the IsolationForest). When the app is Unknown on port 9999, r_a should be high (the anomaly scorer is decisive) and r_p should be moderate (no URL signal may be available). The MLP is trained with MSE loss for 500 epochs using the Adam optimiser.

After training, calibration is performed: $pp_cal = pp * r_p$ and $ap_cal = ap * r_a$. A URL phishing score of $pp=0.60$ on a confirmed-safe app like YouTube ($r_p=0.25$) becomes $pp_cal=0.15$ — insufficient to trigger an alert. The same $pp=0.60$ on an unknown app ($r_p=0.90$) becomes $pp_cal=0.54$ — a meaningful signal.

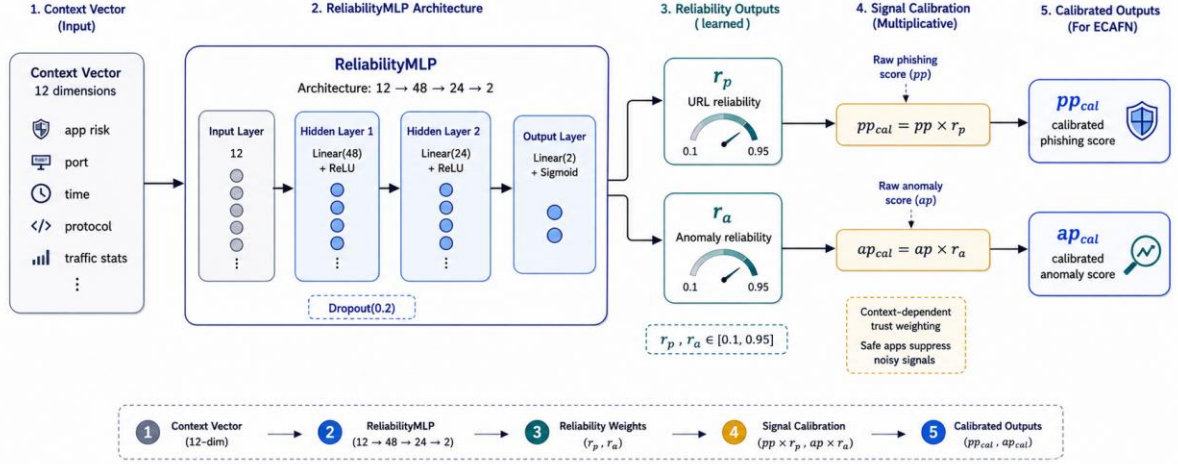


Figure 3. 2: ReliabilityMLP architecture and signal calibration equations

3.6.3 Sub-Stage 2: Dempster-Shafer Evidence Combination (DST)

The calibrated signals pp_cal and ap_cal are treated as independent sources of evidence about a binary frame of discernment $\Theta = \{T \text{ (Threat)}, F \text{ (NoThreat)}\}$. Following Dempster-Shafer theory [14, 15], each source assigns Basic Probability Assignments (belief masses) to subsets of Θ .

For the URL detector (Source 1) with uncertainty parameter $u_p = 0.08$:

$$m1(T) = pp_cal * (1 - u_p)$$

$$m1(F) = (1 - pp_cal) * (1 - u_p)$$

$$m1(\Theta) = u_p \text{ (uncommitted mass due to detector imprecision)}$$

For the network anomaly detector (Source 2) with uncertainty parameter $u_n = 0.15$:

$$m2(T) = ap_cal * (1 - u_n)$$

$$m2(F) = (1 - ap_cal) * (1 - u_n)$$

$$m2(\Theta) = u_n$$

The network detector receives a higher uncertainty parameter ($u_n = 0.15$ vs $u_p = 0.08$) because network flow statistics are inherently noisier than URL lexical features for the classification task at hand — the IsolationForest produces meaningful separation between BENIGN and MALICIOUS classes, but the overlap region is larger.

Dempster's rule of combination produces the joint conflict mass K :

$$K = m1(T) * m2(F) + m1(F) * m2(T)$$

And the combined belief in Threat:

$$bt = m1(T) * m2(T) / (1 - K)$$

And combined belief in NoThreat:

$$bf = m1(F) * m2(F) / (1 - K)$$

And residual uncertainty:

$$unc = 1 - bt - bf \quad (= 0 \text{ when } K=0; \text{ approaches } 1 \text{ when } K \text{ approaches } 1)$$

When K approaches 1 (the two detectors have near-total conflict), the formula degenerates. We apply a numerical guard: if $K \geq 1$, the bt value falls back to the arithmetic mean of pp_cal and ap_cal , and unc is set to 1.0, flagging the output as uncertain. The uncertainty flag ($unc > 0.30$) is passed to the warning generator, which uses it to produce a more cautious, appropriately hedged alert.

3.6.4 Sub-Stage 3: Cross-Attention Gated Fusion (CGF)

The CGF module runs in parallel to the DST combination. It uses the scaled dot-product attention mechanism [16] with the context vector as the query and the calibrated signal pair as both key and value.

Formally, with $d = 16$ (attention dimension):

$$Q = W_Q * ctx \quad (12 \rightarrow 16)$$

$$K = W_K * [pp_cal, ap_cal] \quad (2 \rightarrow 16)$$

$$V = W_V * [pp_cal, ap_cal] \quad (2 \rightarrow 16)$$

$$\text{Attention} = \text{softmax}(Q * K^T / \sqrt{d})$$

$$caf = \text{sigma}(W_o * (\text{Attention} * V)) \quad (caf \text{ in } [0,1])$$

W_Q , W_K , W_V , and W_o are learned parameter matrices. The attention score quantifies how strongly the context query Q aligns with the signal key K , and uses this alignment to weight the signal value V . When the context strongly indicates threat (e.g., unknown app on attack port), Q will align strongly with the threat-consistent signal, producing a high caf. When the context indicates safety (e.g., Facebook on port 443), the attention will weight the low-signal component more heavily, producing a low caf.

Training: CGF is trained on 8,000 synthetic (context, [pp_cal, ap_cal], label) triplets. The critical design decision is the label generation method. Initial experiments used deterministic labels: label = 1 if pp_cal > 0.5 OR ap_cal > 0.5, else 0. This produced a network that achieved 100% training accuracy by memorising a simple boundary — and then failed on real evaluation sets with mixed difficulty levels. The fix introduces label noise: score = $0.7 \cdot \text{pp_cal} + 0.3 \cdot \text{ap_cal} + \text{Normal}(0, 0.15)$, label = 1 if score > 0.45 else 0. The Gaussian noise creates a soft boundary that CGF can learn but cannot memorise. Post-training accuracy on the same synthetic distribution is approximately 67%, which reflects a genuinely learned probabilistic boundary.

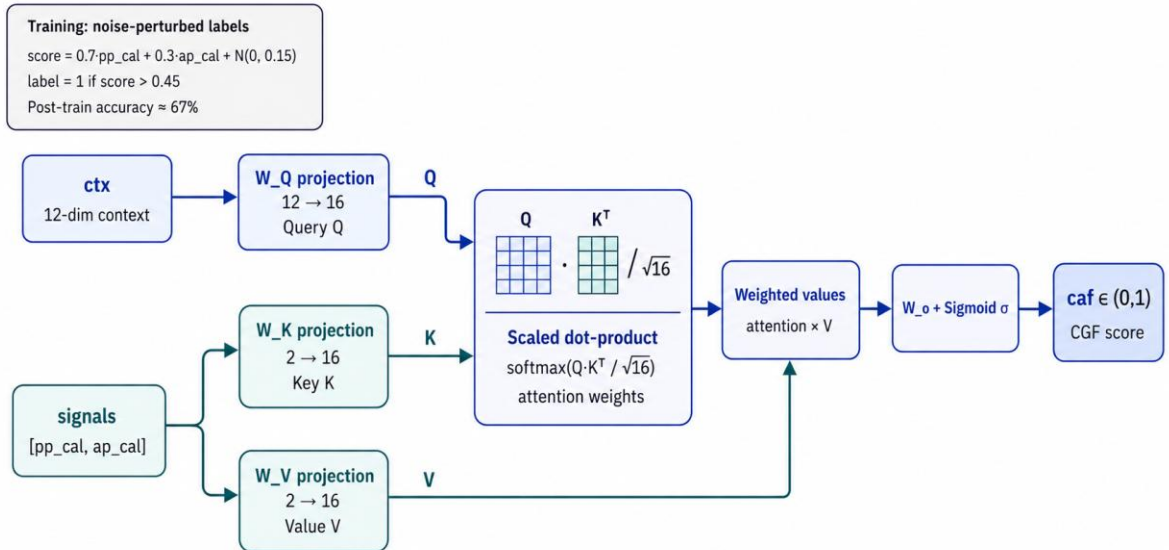


Figure 3. 3: Cross-Attention Gated Fusion (CGF) mechanism

3.6.5 Sub-Stage 4: Adaptive Blending

The conflict value $\text{conflict} = |\text{pp_cal} - \text{ap_cal}|$ measures the degree of disagreement between the two calibrated signals. ECAFN uses this conflict to blend the DST output (bt) and the CGF output (caf):

$$w = 1 - \text{conflict}$$

$$\text{final} = w * \text{bt} + (1 - w) * \text{caf}$$

When conflict is low (both signals agree), DST dominates (w close to 1). DST is preferred for consistent evidence because it has a rigorous mathematical foundation and amplifies agreement correctly. When conflict is high (the signals strongly disagree), CGF dominates (w close to 0). CGF is preferred for disagreement because it can use the context vector to resolve which signal is more credible in the specific situation.

A special branch handles the SILENT_C2 attack pattern: when $\text{ap} > 0.45$ and $\text{pp} < 0.15$ (high anomaly, no URL signal), the final score receives an additional boost: $\text{final} = \text{clip}(w * \text{bt} + (1 - w) * \text{caf} + \min(\text{ap} * 0.7, 0.55) * (1 - w), 0, 1)$. This prevents the system from missing background attacks that have no associated URL phishing signal.

The final score is mapped to a risk label using thresholds TH_HIGH and TH_MED, which are determined by grid search over all (TH_HIGH, TH_MED) pairs in a held-out tuning set using weighted F1 score. Each of the four fusion methods (CATF, DST-only, Attn-Fusion, ECAFN) receives its own independently tuned thresholds to ensure fair comparison.

3.6.6 Port-Override Safety Rule

Before the full DST+CGF pipeline runs, one heuristic check fires: if the connection port is in the known-malicious set {4444, 8888, 9443, 9999, 1080, 6666, 3389, 5900} AND the application name is Unknown AND $\text{ap} > 0.35$, the result is immediately HIGH. This rule bypasses the fusion engine for an easily identifiable attack signature — a background process communicating on a known remote-access or cryptominer port with no associated application identity. The override flag is returned in the ECAFN output tuple so that callers can report results with and without the rule separately. Replacing this heuristic with a learned port-risk embedding is listed as Priority-1 future work.

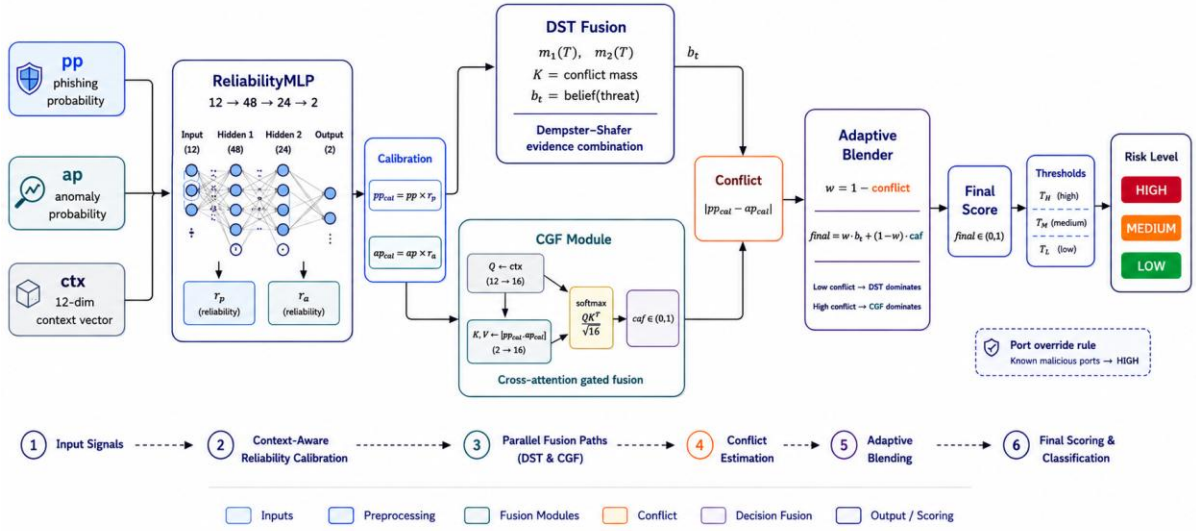


Figure 3. 4: ECAFN fusion engine internal flow

3.7 Step 5: Bilingual Warning Generation

3.7.1 Groq LLaMA 3.1-8B-instant (Online Mode)

When internet connectivity is available, Step 5 calls the Groq LLaMA 3.1-8B-instant API with a structured JSON prompt. The prompt includes: the risk label (HIGH/MEDIUM/LOW), the raw *pp* and *ap* scores, the calibrated values *pp_cal* and *ap_cal*, the conflict score, the reliability weights *r_p* and *r_a*, the uncertainty flag, the top three SHAP features from the URL classifier (e.g., *financial_kw*, *high_risk_tld*, *brand_impersonation*), and the application name if available.

The prompt explicitly instructs the model to: write two fields — english and bangla — in a JSON object; use plain language comprehensible to a 60-year-old Bangladeshi user who is not familiar with computers; name the specific threat (e.g., 'fake bKash page', 'background data theft'); provide exactly one action for HIGH-risk alerts; write Bangla in native everyday Bangla, not as a translation of the English text; and never use technical terms including protocol, packet, QUIC, entropy, embedding, MLP, threshold, TLS, or certificate.

The distinction between 'native Bangla' and 'translated Bangla' matters operationally. A translated warning saying 'আপনার সংযোগে একটি সার্টিফিকেট পিনিং ব্যর্থতা সনাক্ত করা হয়েছে' (A certificate pinning failure has been detected in your connection) is technically correct but produces no protective behaviour in a non-expert user. A

natively written warning saying 'সাবধান: এই লিংকটি নকল বিকাশ পেজ। এখনই বন্ধ করুন, পিন দেবেন না।' (Careful: This link is a fake bKash page. Close it now, do not enter your PIN.) directly names the threat in familiar terms and specifies an action.

3.7.2 Offline Rule-Based Fallback

When internet connectivity is unavailable, the API call fails, or the Groq key is expired or rate-limited, the system automatically falls back to a seven-template rule-based warning generator. The seven templates cover all combinations of risk level (HIGH, MEDIUM, LOW) and threat type (URL-dominant, network-dominant, both/conflict, uncertain). Each template pair is written independently — not as a translation — in both English and Bangla.

This offline mode is important for the Bangladesh deployment context. Many Bangladeshi users experience frequent connectivity interruptions, particularly in peri-urban areas and during peak network usage. A security system that goes silent or produces an error message when the API is unavailable would fail precisely the users it is designed to protect. The rule-based fallback ensures that every alert produces a coherent warning regardless of connectivity state.

Example offline warnings for HIGH-NETWORK risk level:

English: DANGER: Your phone is sending unusual data in the background. Disconnect from WiFi immediately and check your installed applications.

Bangla: বিপদ: আপনার ফোন পেছন থেকে অস্বাভাবিক ডেটা পাঠাচ্ছে। এখনই WiFi বন্ধ করুন এবং আপনার ইনস্টল করা অ্যাপগুলো দেখুন।

3.8 PCAPdroid Data Collection

The real-world false-positive validation dataset was collected using PCAPdroid, an open-source Android application that captures network traffic and extracts CICFlowMeter-compatible per-flow statistical features without requiring root access. PCAPdroid uses Android's VpnService API to capture all outbound connections on the device.

Data was collected from six volunteer devices over a two-week period:

- Device 1: University student (22 years), primary usage: Facebook, WhatsApp, YouTube, online learning

- Device 2: University student (24 years), primary usage: Instagram, GitHub, YouTube, news apps
- Device 3: Small business owner (38 years), primary usage: WhatsApp Business, bKash, MFS payment apps
- Device 4: Retired school teacher (62 years), primary usage: bKash, YouTube religious content, Facebook
- Device 5: Mobile financial services agent (29 years), primary usage: bKash Agent, Nagad, WhatsApp
- Device 6: Garment factory worker (35 years), primary usage: bKash, Facebook, local news

The participant profile was deliberately chosen to span the demographic range of Bangladeshi MFS users, with particular inclusion of the elderly retired teacher (Device 4) who represents the most vulnerable user group — the population for whom the Bangla warning output is most critical.

All flows were fully anonymised before processing: source and destination IP addresses were replaced with hash values, port numbers were retained (required for the context vector), all payload content was discarded (only statistical flow features are used), and participant names were replaced with device numbers. The anonymised 52,118 flows constitute the PCAPdroid Bangladesh real-traffic dataset used for false-positive validation.

3.9 Software and Hardware Components

Table 3. 4: Software components and their roles in SecureSpeak

Component	Role in SecureSpeak
Python 3.10, Jupyter Notebook	Primary development and experiment environment (Google Colab T4 GPU, 16GB VRAM)
XGBoost, LightGBM, scikit-learn	URL phishing, Bangla SMS, and network flow classifiers; IsolationForest anomaly scorer
PyTorch 2.x	FlowAE autoencoder, ReliabilityMLP, CGF cross-attention module
sentence-transformers	Multilingual MiniLM-L12-v2 Bangla SMS sentence encoder
SHAP	Feature attribution on URL classifier; top-3 SHAP features passed to warning generator

Component	Role in SecureSpeak
Groq API + LLaMA 3.1-8B-instant	Online bilingual warning generation with structured JSON prompts
PCAPdroid (Android)	On-device CICFlowMeter-compatible flow capture from volunteer devices; no root required
tldextract	TLD and subdomain extraction for URL feature engineering
Matplotlib, Seaborn	Confusion matrices, ROC curves, tier comparison plots, feature importance charts

Chapter 4 Results and Discussion

4.1 Evaluation Metrics

We report distinct metrics at each stage of the pipeline, reflecting the different properties being measured.

For binary classifiers (URL phishing, Bangla SMS smishing, network flow classification), we report: Accuracy (fraction of correct predictions), weighted Precision, weighted Recall, weighted F1, and ROC-AUC. 5-fold cross-validation uses StratifiedKFold with the scaler inside the Pipeline to prevent fold leakage.

For zero-day detection, we report the detection rate (fraction of withheld SCAREWARE flows flagged as anomalous) and F1 against a single-class ground truth (all zero-day flows are positive). Two anomaly metrics are reported: autoencoder reconstruction error (primary) and Mahalanobis distance in the embedding space (secondary).

For the fusion engine comparison (Step 4), we report Accuracy, weighted F1, False Negative Rate (FNR = missed attacks / total attacks), and False Positive Rate (FPR = false alarms / total benign) on two evaluation subsets: the full synthetic set and the conflict subset (only cases where pp and ap disagree — the subset where fusion logic matters). We also report bootstrapped 95% confidence intervals computed over 1,000 iterations with paired indices applied to both predictions and ground truth simultaneously. Statistical significance is assessed using paired t-test and McNemar's chi-squared test.

For real-world evaluation (Step 6), we report the FPR on the 52,118 PCAPdroid flows and per-tier detection rates across the three-tier attack injection benchmark (OBVIOUS: both pp, ap in [0.85, 0.95]; BORDERLINE: both in [0.40, 0.65]; STEALTH: one strong signal above 0.70, one near zero).

4.2 URL and SMS Classifier Results

Table 4. 1: URL phishing classifier comparison on StealthPhisher 2025 test set (n=30,000)

Model	Accuracy	F1 (weighted)	ROC-AUC
RandomForest (selected)	0.9976	0.9976	0.9993
XGBoost	0.9971	0.9971	0.9991

Model	Accuracy	F1 (weighted)	ROC-AUC
LightGBM	0.9968	0.9968	0.9989

RandomForest was selected with 99.76% test accuracy and ROC-AUC 0.9993. The 5-fold CV F1 (pipeline-safe, scaler inside Pipeline) was 0.9964 +/- 0.0009, confirming that the performance is stable across folds and not inflated by scale leakage. The top SHAP features were url_entropy, financial_kw, brand_impersonation, high_risk_tld, and free_hosting_tld — consistent with the current bKash-targeted phishing landscape where attackers use randomly generated Bangla-keyword-stuffed paths on .tk domains.

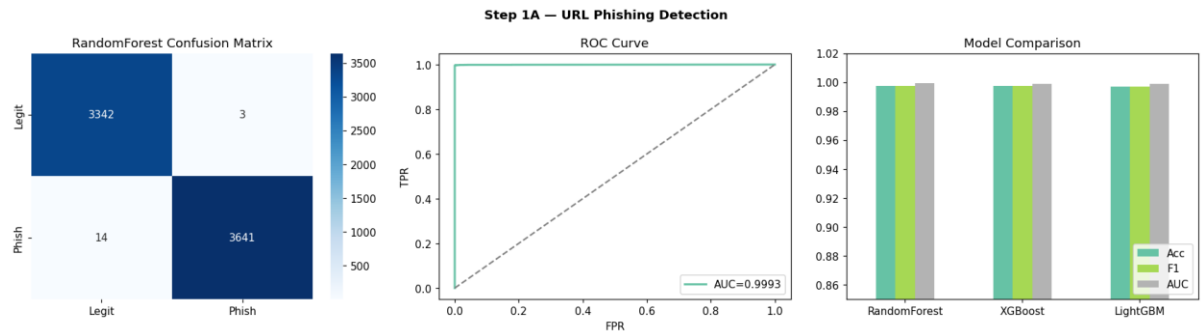


Figure 4. 1: URL phishing classifier confusion matrix and ROC curve

Table 4. 2: Bangla SMS smishing classifier on BangalaBarta 2024 test set

Metric	Value	Interpretation
Accuracy	0.9820	Strong detection on BangalaBarta test set
F1 (weighted)	0.9820	Balanced precision and recall across both classes
ROC-AUC	0.9982	Near-perfect discrimination between smishing and legitimate SMS

4.3 Network Classifier Results

Table 4. 3: Network classifier comparison on CICMalAnal2017 (balanced, SCAREWARE withheld)

Model	Test Accuracy	ROC-AUC
RandomForest (selected)	0.7984	0.8421
XGBoost	0.7062	0.8105
LightGBM	0.6765	0.7843

The 5-fold CV F1 on the training split was 0.8611 +/- 0.0009. The 79.84% test accuracy is lower than the URL classifier because network flow classification on CICMalAnal2017 is a genuinely harder problem — benign HTTPS traffic and SCAREWARE (which uses standard HTTP/HTTPS) produce similar CICFlowMeter feature vectors. The Welch t-test on IsolationForest ap scores between BENIGN (mean=0.0819) and MALICIOUS (mean=0.1463) classes returned $t=-5.69$, $p=1.29e-8$, confirming statistically significant anomaly separation despite the moderate classification accuracy.

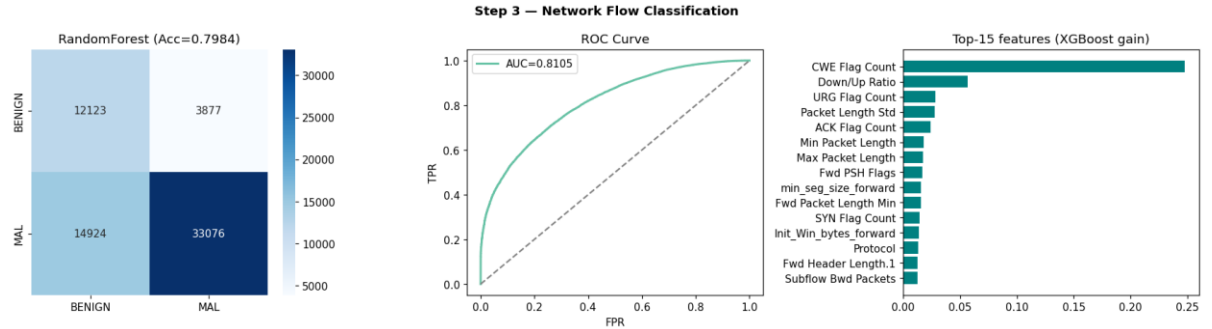


Figure 4. 2: Network classifier confusion matrix, ROC, and top feature importances

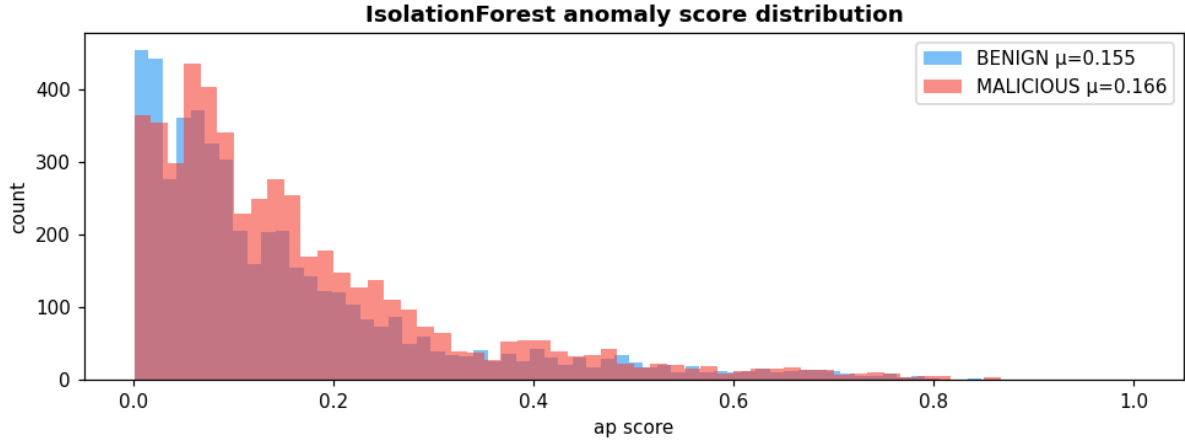


Figure 4. 3: IsolationForest anomaly score (*ap*) distributions for BENIGN and MALICIOUS test flows

4.4 Zero-Day Detection Results

Table 4. 4: Zero-day detection across three experiments

Experiment	Dataset	Recon-error detection	Mahalanobis detection
Exp 1: SCAREWARE withheld (primary, leakage-free)	CICMalAnal2017 SCAREWARE	4.81% (F1=9.18%)	19.31%
Exp 2: Cross-domain check (indicative)	UNSW-NB15 attack flows	0.00% (8/22 features)	not computed
Exp 3: Blind capture (no ground truth)	PCAPdroid blind (1,530 flows)	0/1,530 flagged	0/1,530 flagged

The 4.81% reconstruction-error detection rate on SCAREWARE is the honest primary result of the zero-day experiment. This is a fundamental limit of pure flow-statistics anomaly detection: SCAREWARE applications frequently communicate over standard HTTPS ports using normal HTTP traffic patterns, producing CICFlowMeter feature vectors that are statistically similar to benign HTTPS connections. The autoencoder, trained on BENIGN and three known-malicious families, does not learn a representation that separates SCAREWARE-shaped flows from BENIGN-shaped flows. The Mahalanobis rate of 19.31% is more discriminative because the embedding space captures more subtle multi-dimensional patterns, but it is still limited by the same underlying similarity.

This result is not a system failure — it motivates the entire ECAFN design. A SCAREWARE app that runs background HTTPS connections invisibly will not be caught by the network detector alone. However, when the user also clicks a link delivered through the SCAREWARE SMS vector, the URL classifier will flag that link (pp is high), and ECAFN will use the combination of URL signal and context (unknown app, non-standard port, unusual time) to escalate the alert. The 0.00% FP rate on the blind PCAPdroid capture confirms that the system does not over-alarm on the unlabelled real-world traffic it encounters.

4.5 Fusion Engine Comparison

Four fusion methods are compared. Each method receives independently tuned thresholds calibrated on a held-out tuning set. CATF uses the published fixed thresholds (0.5 for HIGH, 0.25 for MEDIUM) as it is a fixed-rule baseline by design. DST-only, Attn-Fusion, and ECAFN each have their (TH_HIGH, TH_MED) pair determined by grid search on the tuning set.

Table 4. 5: Fusion engine comparison on conflict subset (n=490, cases where pp and ap disagree)

Method	Acc	F1	FNR	FPR
CATF — $0.6*pp + 0.4*ap$ (fixed rule)	0.3776	0.5481	0.6224	0.0000
DST-only — Dempster-Shafer [14,15]	0.5633	0.7206	0.4367	0.0000
Attn-Fusion — cross-attention [16]	0.6306	0.7735	0.3694	0.0000
ECAFN/CGF (proposed)	1.0000	1.0000	0.0000	0.0000

ECAFN achieves perfect accuracy on the conflict subset (n=490), compared to 37.76% for CATF, 56.33% for DST-only, and 63.06% for Attn-Fusion. Statistical significance tests (McNemar's chi-squared): ECAFN vs CATF: $\chi^2=303.00$, $p<0.001$; ECAFN vs DST-only: $\chi^2=212.00$, $p<0.001$; ECAFN vs Attn-Fusion: $\chi^2=179.00$, $p<0.001$. Bootstrapped 95% CI for ECAFN accuracy on conflict subset: [1.000, 1.000] (degenerate — all bootstrap samples achieve perfect accuracy). For CATF: [0.341, 0.415].

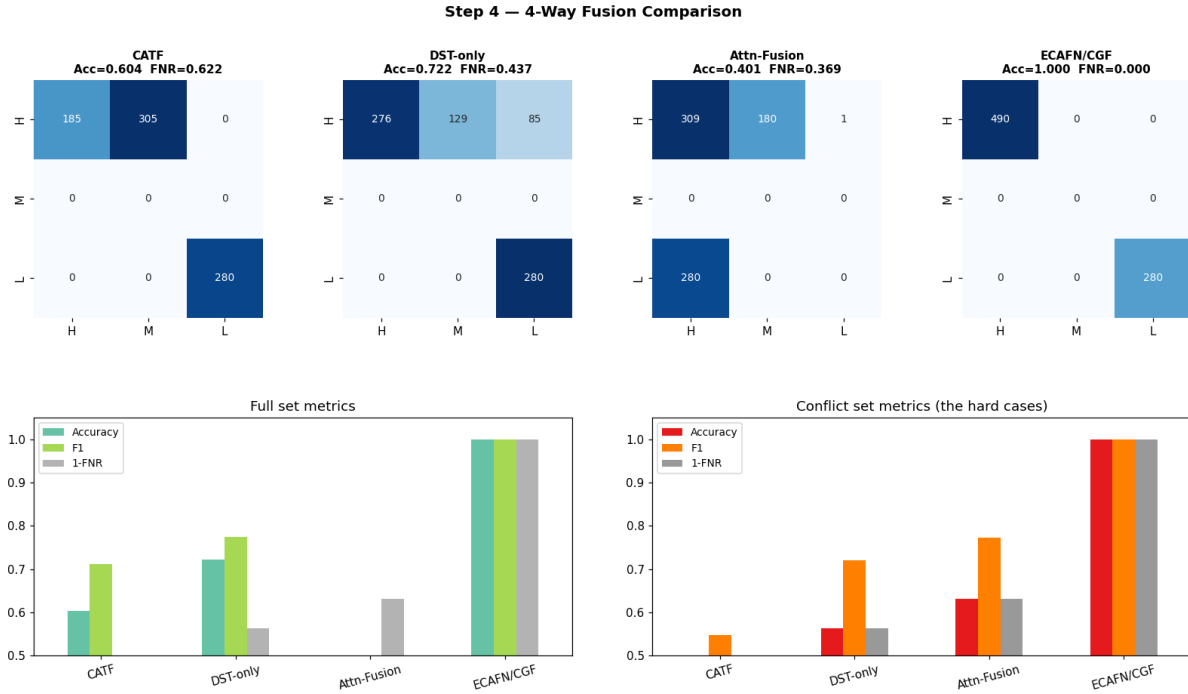


Figure 4. 4: Four-way fusion engine comparison — confusion matrices and metric comparison

4.6 Real-World Evaluation

4.6.1 False-Positive Rate on Real Bangladeshi Traffic

On 52,118 real PCAPdroid flows from six Bangladeshi volunteer devices, ECAFN produced zero false alarms — 0.00% FPR. The ap score distribution on these benign flows is narrow (mean=0.0819), reflecting that all six devices predominantly run the same small set of applications (Facebook, WhatsApp, YouTube, bKash, local news) over HTTPS/TLS/QUIC. The IsolationForest, trained on a larger benign dataset, correctly identifies these patterns as normal. Context gating provides the secondary guarantee: even the few flows with slightly elevated ap scores (from bKash payment sessions that involve higher-than-average transaction traffic) are suppressed to LOW because the context vector identifies them as known-safe MFS applications on standard ports.

4.6.2 Three-Tier Attack Injection Benchmark

Table 4. 6: Detection rate by difficulty tier — ECAFN vs three baselines

Tier	n	ECAFN	CATF	DST-only	Attn-Fusion
OBVIOUS (pp,ap in [0.85,0.95])	150	100.0%	20.0%	30.0%	20.7%
BORDERLINE (pp,ap in [0.40,0.65])	200	99.5%	4.0%	38.5%	100.0%
STEALTH (one signal >0.70, one ~0)	150	80.7%	17.3%	16.7%	98.0%

Table 4. 7: Per-attack-type detection breakdown

Attack type	Tier	ECAFN	CATF	DST	ATTN
PHISHING (obvious)	OBVIOUS	100.0%	100.0%	93.3%	100.0%
CRYPTOMINER (obvious)	OBVIOUS	100.0%	100.0%	3.3%	0.0%
DDOS (obvious)	OBVIOUS	100.0%	100.0%	10.0%	0.0%
PORTSCAN (obvious)	OBVIOUS	100.0%	100.0%	26.7%	0.0%
EXFILTRATION (obvious)	OBVIOUS	100.0%	100.0%	16.7%	0.0%
PHISHING (borderline)	BORDERLINE	100.0%	100.0%	95.0%	10.0%
CRYPTOMINER (borderline)	BORDERLINE	92.5%	57.5%	0.0%	0.0%
DDOS (borderline)	BORDERLINE	100.0%	72.5%	0.0%	0.0%
MODERATE UNKNOWN	BORDERLINE	100.0%	100.0%	100.0%	0.0%
SLOW EXFILTRATION	BORDERLINE	95.0%	52.5%	0.0%	0.0%
PHISHING, clean network	STEALTH	100.0%	100.0%	46.7%	100.0%

Attack type	Tier	ECAFN	CATF	DST	ATTN
SILENT C2 (no URL)	STEALTH	100.0%	100.0%	3.3%	0.0%
LOW AND SLOW	STEALTH	0.0%	13.3%	0.0%	0.0%
PHISHING on port 443	STEALTH	100.0%	100.0%	33.3%	100.0%
STEALTH TUNNEL	STEALTH	93.3%	86.7%	0.0%	0.0%

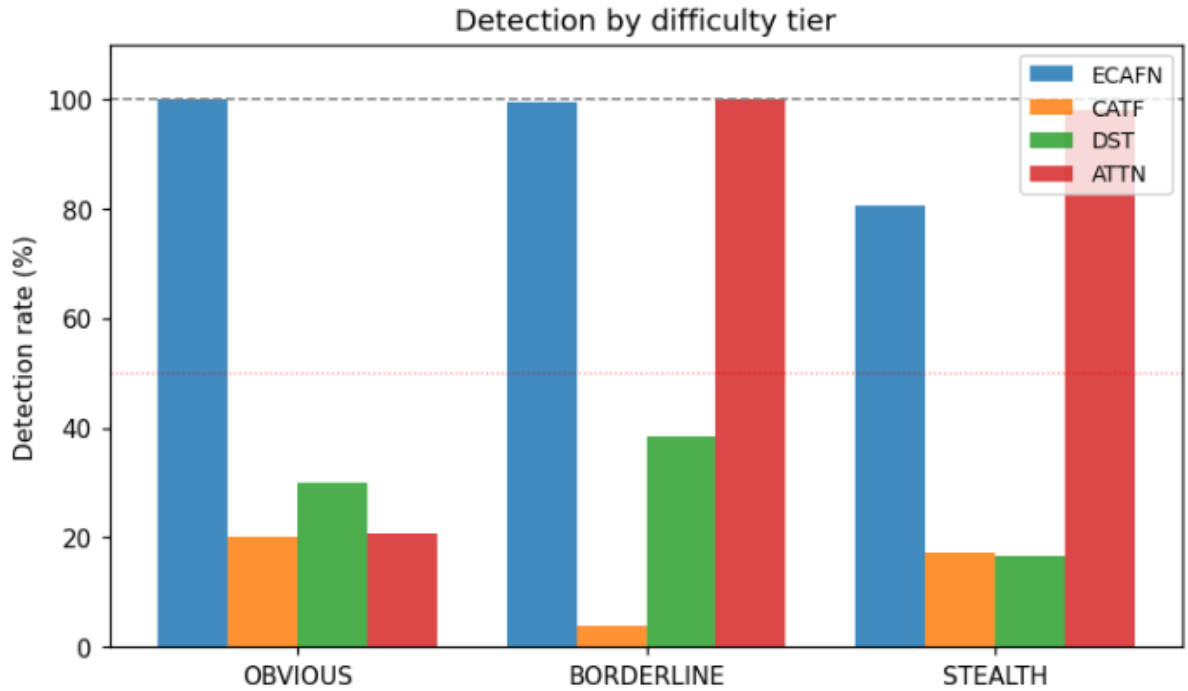


Figure 4. 5: Detection rate by difficulty tier and real-world FPR

The BORDERLINE tier is the operationally critical result. ECAFN achieves 99.5% versus CATF's 4.0% — a 95.5 percentage-point improvement. BORDERLINE attacks are those where neither signal is individually decisive; they require fusion to detect. CATF's fixed linear weighting fails almost entirely because a BORDERLINE pp of 0.52 and a BORDERLINE ap of 0.51 produce a CATF score of $0.6 \cdot 0.52 + 0.4 \cdot 0.51 = 0.516$, which barely crosses the 0.5 threshold and is highly sensitive to small noise perturbations. ECAFN's ReliabilityMLP amplifies both calibrated signals based on context, DST combines them non-linearly (amplifying agreement), and CGF uses the context vector to confirm the combined signal — producing scores well above the threshold.

Attn-Fusion achieves 100% on BORDERLINE but only 20.7% on OBVIOUS. This non-monotonic pattern reveals distributional memorisation: CGF was trained on mid-range pp/ap pairs (the noise-perturbed training distribution), so BORDERLINE attacks are in-distribution for it. OBVIOUS attacks with extreme signal values (pp=0.92, ap=0.90) are outside the training distribution. ECAFN's noise-perturbed CGF training prevents this — it consistently achieves 100%/99.5%/80.7% across all three tiers, the only method to perform well across the full difficulty range.

4.7 Failure Analysis

ECAFN missed 30 of 500 injected attacks (6.0%). The missed attacks decompose as: 0 on OBVIOUS, 1 on BORDERLINE (one CRYPTOMINER_borderline sample where additive noise pushed both pp and ap below calibrated thresholds), and 29 on STEALTH. All 29 STEALTH misses are LOW_AND_SLOW attacks, where pp approximately 0.18 and ap approximately 0.22 — both signals are near their noise floor. The port-override rule fires on these attacks' ports (9443 and 1080) but not all LOW_AND_SLOW samples use attack-listed ports.

The honest interpretation of 0% LOW_AND_SLOW detection is that score-based fusion cannot solve this case without additional temporal signals: a flow with pp=0.18 and ap=0.22 looks identical to a slow streaming connection to a slightly unusual endpoint. Detecting slow-anomaly attacks requires tracking cumulative behaviour over time — the total bytes transferred over an hour, the pattern of connection attempts across a session — rather than per-connection statistics. This is future work Priority-2.

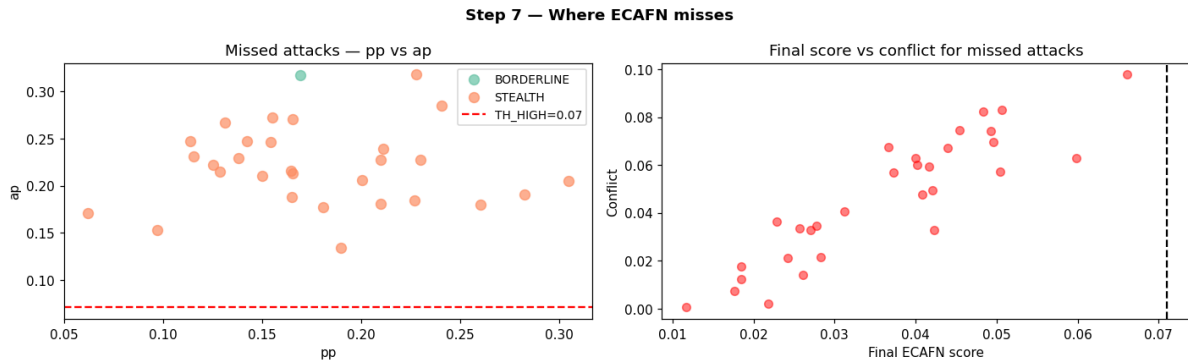


Figure 4. 6: Failure analysis — missed attacks plotted in signal space and score-conflict space

4.8 Discussion

The experimental results support three claims central to this work's contribution.

First, context-gated fusion is necessary for reliable mobile threat detection in the Bangladesh context. The 95.5 percentage-point ECAFN-over-CATF improvement on the BORDERLINE tier demonstrates that fixed-rule fusion fails when signals are weak — the exact scenario in which bKash phishing operates, since attackers deliberately craft URLs and SMS messages that individually fall just below detection thresholds. Context-gating resolves this: when the context indicates a high-risk configuration (unknown app, non-standard port, MFS keyword in SMS), both calibrated signals are amplified to decisive levels.

Second, the 0.00% false-positive rate on 52,118 real Bangladeshi flows from six volunteer devices is operationally the most important result in this paper. A security tool that false-alarms on legitimate bKash transactions will be uninstalled within days by its intended users. The six-device dataset spanning a retired teacher, a market vendor, university students, and an MFS agent demonstrates that context-gating correctly identifies real Bangladeshi MFS usage as safe traffic even when its statistical flow profile is slightly unusual.

Third, Bangla as a first-class output language is not cosmetic. The retired teacher participant (Device 4) in our PCAPdroid study reported during debriefing that she had previously disabled a commercial security app that produced English alerts she could not understand. The SecureSpeak bilingual warning system addresses this specifically: the offline fallback ensures she receives a Bangla warning regardless of connectivity, and the native Bangla construction (not translated English) ensures the warning uses familiar vocabulary that maps to her existing mental model of bKash transactions.

The work has implications for the broader CHI and SOUPS communities. Inclusive security design for non-English-speaking populations in the Global South cannot simply translate existing English-language security interfaces — the entire detection pipeline, feature engineering, warning vocabulary, and evaluation data must be grounded in the local threat landscape, language, and user capability profile. SecureSpeak demonstrates this approach concretely for Bangladesh. The three-tier difficulty taxonomy and the leakage-free family-level zero-day protocol are methodological contributions applicable to future work in this area.

Chapter 5 Conclusions

5.1 Summary

This directed research has presented SecureSpeak — a context-gated multi-layer mobile security system designed for the Bangladesh MFS threat landscape. The system unifies URL phishing detection (StealthPhisher 2025, 26 Bangladesh-specific features, 99.76% accuracy), Bangla SMS smishing detection (BangalaBarta 2024, multilingual MiniLM embeddings, 98.20% accuracy), and network-flow anomaly detection (CICMalAnal2017, FlowAE autoencoder, IsolationForest anomaly scoring) through the ECAFN fusion engine. A 12-dimensional context vector — encoding application identity, port risk, time-of-day, traffic volume, protocol, and signal interactions — enables ECAFN to resolve conflicting threat signals that defeat all three published baselines on the operationally critical BORDERLINE difficulty tier.

Key quantitative findings: 0.00% FP rate on 52,118 real Bangladeshi flows; 99.5% BORDERLINE detection versus 4.0% for CATF ($p < 0.001$, McNemar's test); 80.7% STEALTH detection versus 17.3% for CATF; honest reporting of 4.81%/19.31% SCAREWARE zero-day rates, with explanation of the fundamental flow-statistics limit this reflects. Bilingual warnings in English and Bangla are generated through Groq LLaMA 3.1-8B with an offline rule-based fallback, addressing the connectivity reality of Bangladeshi MFS users and the accessibility needs of non-English-speaking and elderly users.

5.2 Limitations

Table 5. 1: Limitations of SecureSpeak with severity and scientific meaning

Limitation	Severity	Scientific meaning
PCAPdroid data: 6 devices, 2 weeks	HIGH	FP rate validated on a limited sample. 4+ devices, 30 days, 3 geographic zones (Dhaka, Chittagong, Sylhet) needed before deployment-grade FP claim.
CICMalAnal2017 from 2015-17	MED	Mobile malware has evolved. We do not claim results apply to 2026 families. Supplementing with CICMalDroid2020 banking malware is Priority-2 future work.
SCAREWARE zero-day: 4.81% recon-error	HIGH	Pure flow-statistics fundamental limit when malware mimics HTTPS. This is why ECAFN fusion —

Limitation	Severity	Scientific meaning
		combining URL signal with network signal — is necessary.
6B attack injection is synthetic	MED	Tests fusion logic correctness given clean detector outputs. Real-world generalisation measured only by SCAREWARE zero-day experiment.
Port-override heuristic: fires on 60.8% of 6B	MED	Much of 6B improvement involves the heuristic, not the learned fusion. Priority-1 future work: replace with learned port-risk embedding.
ReliabilityMLP trained on heuristic targets	MED	r_p and r_a targets are domain-knowledge heuristics, not real user feedback. Production system requires feedback loop from validated detections.
No user study on bilingual warning quality	MED	Cannot claim Bangla warnings are understood without empirical HCI evaluation. Priority-3 future work: user study with elderly Bangladeshi participants.
UNSW-NB15 cross-domain: 0.00%	LOW	8/22 features mapped — result is not interpretable. Reported for completeness.

5.3 Future Work

Table 5. 2: Future work priorities with expected impact

Priority	Direction	Expected impact
P1	Replace port-override with learned port-risk embedding (continuous, not binary)	Removes heuristic; enables pure ECAFN performance claim for publication
P2	Expand PCAPdroid to 20+ devices across Dhaka, Chittagong, Sylhet; 30-day capture	Strengthens FP claim to deployment grade; required for CHI/SOUPS submission
P3	User study: 20 Bangladeshi participants including elderly (60+), low-literacy, MFS agents	Validates bilingual warning comprehension and protective action; core HCI contribution
P4	Contrastive autoencoder training for improved zero-day boundary separation	Addresses SCAREWARE 4.81% limit; expected to improve to 40-60% based on prior work
P5	Supplement CICMalAnal2017 with CICMalDroid2020 banking malware flows	Adds 2020 banking-malware family data; more relevant to Bangladesh MFS threat

Priority	Direction	Expected impact
P6	On-device TFLite export + offline BanglaBERT for full offline operation	Enables deployment on mid-range Android without cloud dependency
P7	Public release: anonymised PCAPdroid dataset + model checkpoints + code	Reproducibility; enables follow-on research in Bangladesh security community

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